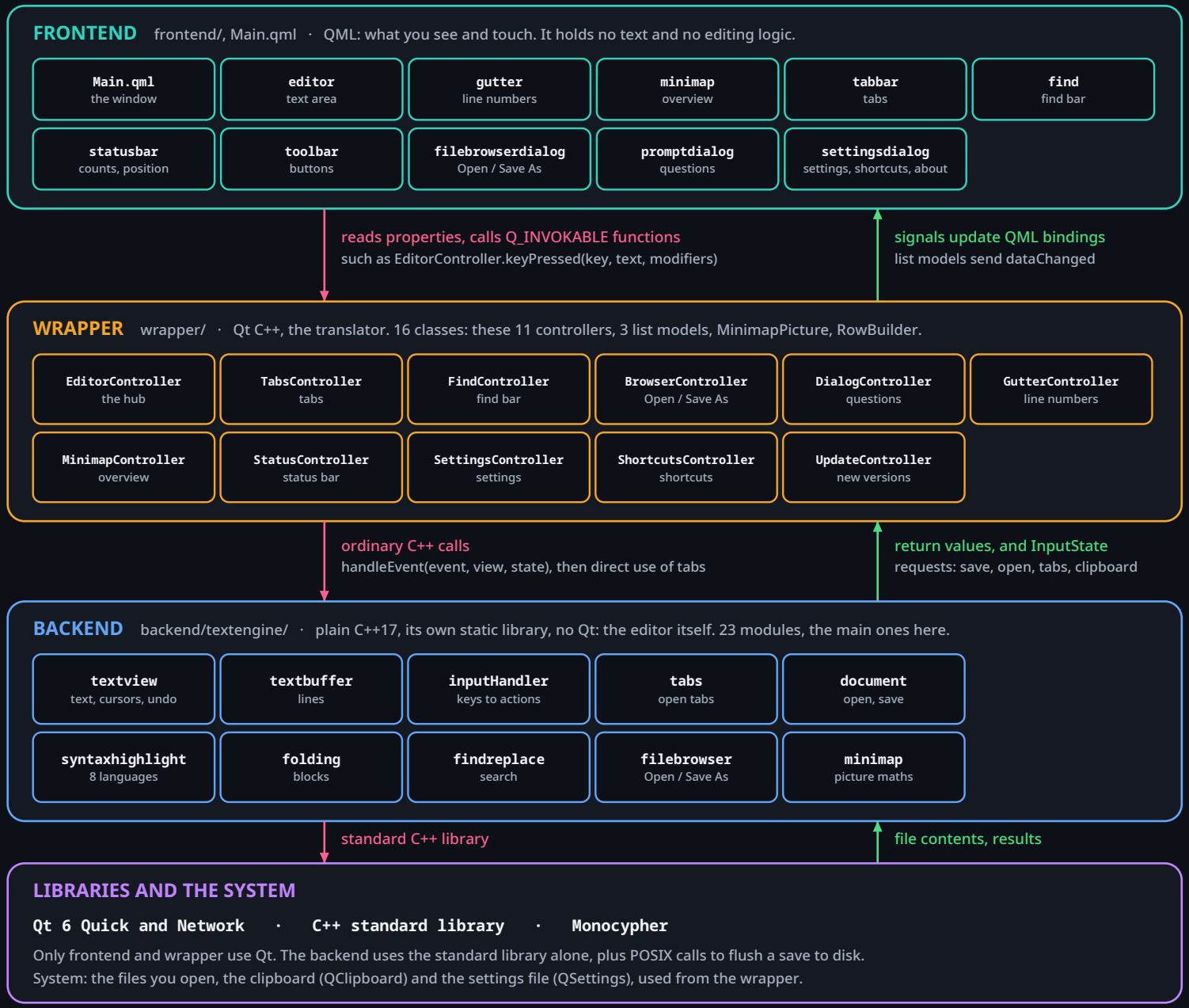
The big picture

Three layers, each talking only to the next one. All of them run on one thread: the app starts no threads of its own.



The rules that keep it clean

The backend knows nothing about Qt.
No backend file includes a Qt header. It also knows no screen: keys, clicks and rows are its own plain types.

QML never reaches the backend.
Every screen goes through a wrapper singleton, such as EditorController.

The engine asks, the wrapper does.
The input handler cannot save or change tabs itself: it adds Requests to InputState, and EditorController carries them out.

One object owns the editor.
EditorController holds the tabs and input state. The other controllers reach them through it.

One thread.
No code starts a thread, so slow work blocks typing. Files over 50 MB are refused.

RowBuilder has no Qt either.
It lives in wrapper/ but is plain C++: Qt classes only copy its results.

What is what

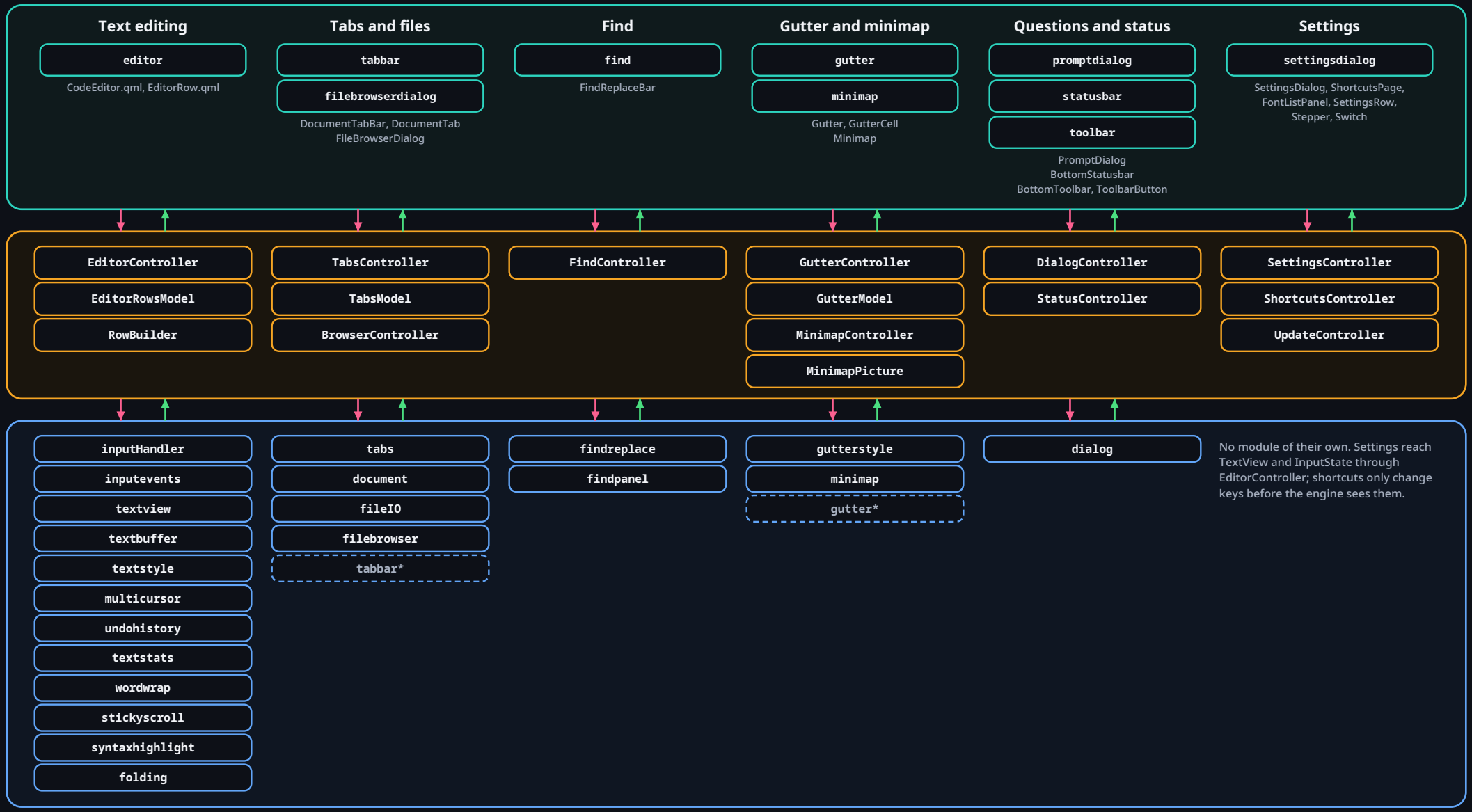
QML	language that describes screens
Controller	C++ class QML can talk to
Singleton	one shared object, used by name
Signal	a class says something changed
Binding	QML value that updates by itself
Q_INVOKABLE	C++ function QML may call
Request	engine asks the program to act

In numbers

QML files (with Main.qml)	**22**
wrapper C++ files, classes	**30, 16**
backend C++ files, modules	**62, 23**

The blueprint

Read each column top to bottom: the screens, the wrapper classes they use, the backend modules behind them. Every part is here once.



calls into the layer below

reports back

`name*` built but not used by the app: nothing includes gutter.h or tabbar.h.

The columns group parts by subject. Classes also use each other, and shared modules reach across columns: page 3 lists them class by class. Above the columns: Main.qml holds all the screens, and every screen shares Theme.qml and Icon.qml.

The connections

Generated from the code: QML names, and the engine and wrapper names used inside each class, checked against its #include chain.

Wrapper class	Used by these screens	Backend modules it uses	Other wrapper parts it uses
EditorController	editor, find, Main	document, fileIO, folding, gutterstyle, inputevents, inputHandler, multicursor, stickyscroll, syntaxhighlight, tabs, textbuffer, textview	DialogController, EditorRowsModel, RowBuilder, ShortcutsController
EditorRowsModel	editor	tabs	RowBuilder
RowBuilder	none: used from C++ only	findreplace, folding, gutterstyle, multicursor, stickyscroll, syntaxhighlight, tabs, textbuffer, textstyle, textview, wordwrap	-
BrowserController	filebrowserdialog, Main	filebrowser, inputevents, inputHandler	EditorController
DialogController	Main, promptdialog	dialog, document, fileIO, inputevents, inputHandler, tabs, textview	EditorController
FindController	find, Main	findpanel, findreplace, inputHandler, textview	EditorController
GutterController	gutter	-	EditorController, GutterModel, RowBuilder
GutterModel	gutter	tabs	RowBuilder
MinimapController	minimap	gutterstyle, inputHandler, minimap, syntaxhighlight, tabs, textview	EditorController, RowBuilder
MinimapPicture	minimap	minimap	MinimapController
SettingsController	editor, settingsdialog	document, inputHandler, tabs, textview	EditorController
ShortcutsController	settingsdialog	-	-
StatusController	Main	document, inputHandler, syntaxhighlight, tabs, textbuffer, textstats, textview	EditorController
UpdateController	settingsdialog, toolbar	document, tabs	DialogController, EditorController
TabsController	Main	inputHandler, tabs	EditorController, TabsModel
TabsModel	Main	-	-

How to read a row: FindController is used by frontend/find and by Main (the toolbar's Find button). It uses the backend modules named, and works through EditorController. "Main" is Main.qml; the other screen names are folders of frontend/.

EditorRowsModel, GutterModel and TabsModel have no QML name: they reach QML through EditorController.rows, GutterController.rows and TabsController.tabs. RowBuilder is used by C++ alone. Calls inside the backend are not listed. ShortcutsController is the only controller that uses no other wrapper class and no backend module.

Following one keystroke

You press the letter a in the text. Everything up to the screen update runs on the GUI thread, one step after the other.

GUI THREAD

all of this project's code, in order

1. **CodeEditor.qml**
   Keys.onPressed gets the key, calls EditorController.keyPressed()
2. **EditorController::keyPressed**
   asks ShortcutsController::remap(), a typed key: makeText("a"), then feed(event, true)
3. **EditorController::feed**
   copies clipboard and file name into InputState, then calls handleEvent(event, view, state)
4. **inputHandler**
   handleEvent > handleText > typeText: mode is Edit, so view.insertText("a")
5. **TextView::applyEdits**
   undo step, TextBuffer::insert, TextStyle::insert, then folding and find matches updated
6. **back in feed**
   carries out the Requests (none for typing), sets the status text, then calls rebuild(true)
7. **EditorController::rebuild**
   updates gutter marks and syntax, sticky regions, wrap width, then RowBuilder::update(tab, width)
8. **EditorRowsModel::refresh**
   dataChanged: QML asks for visible rows again; RowBuilder::row() builds each one when asked
9. **signals**
   stickyChanged, viewChanged, rowsRebuilt, and scrollRequested if the cursor left the screen
10. **documentChanged**
    last signal of feed: every controller below updates itself and QML bindings follow
11. **QML follows**
    the rows changed at step 8; status bar, tabs, find counter and Undo button follow their properties
12. **MinimapController::update**
    runs from a zero timer, so many changes count once; emits changed, MinimapPicture asks for a repaint

QT'S RENDER STEP

Qt's code, not this project's

13. Qt draws the frame. Qt's documentation says MinimapPicture::paint() is called on the renderer thread while the GUI thread is blocked, so it can read the minimap safely. That is the only place another thread could touch this code. In the software test run here, paint() ran on the GUI thread.

Who listens to EditorController

every connect() in the wrapper, plus QML. Nothing calls these one by one: each class connects itself.

documentChanged

- **TabsController::sync**
  titles, unsaved dots, current tab
- **StatusController::sync**
  position, counts, message
- **FindController::sync**
  bar state, counter
- **BrowserController::sync**
  file list

documentChanged, continued

- **DialogController::advance**
  finishes save, then close or quit
- **SettingsController**
  indent for new tabs, Alt+M memory
- **MinimapController::schedule**
  repaint later
- **QML: canUndo, canRedo**
  Undo and Redo buttons

other signals

- **rowsRebuilt: GutterController**
  refresh; Minimap schedule
- **stickyChanged: GutterController**
  refreshSticky; CodeEditor.qml's pinned rows
- **viewChanged: MinimapController**
  schedule
- **scrollRequested: CodeEditor.qml**
  scrolls the list

Starting up

main.cpp is short: it starts Qt and loads Main.qml. The controllers are not made in main.cpp. QML creates each one the first time it uses it.

0 Before main()
each of the 8 language headers registers its LanguageDef (a static variable)

→ **1 QGuiApplication**
Qt starts

→ **2 QQmlApplicationEngine**
if Main.qml fails to load, objectCreationFailed exits with -1

→ **3 loadFromModule(...)**
"ChaoticNotepad", "Main": this creates the window and every screen

During step 3: the order each constructor started, measured

on Qt 6.11.2. Qt decides this order; only the arrows lower on this page are fixed by the code.

1. `MinimapPicture`
2. `MinimapController`
3. `EditorController`
4. `DialogController`
5. `BrowserController`
6. `ShortcutsController`
7. `SettingsController`
8. `StatusController`
9. `FindController`
10. `GutterController`
11. `TabsController`

Models, RowBuilder and the tabs are made inside their owners.

4 Read the command line
each argument that does not start with "-" becomes a file URL, relative names counted from the current folder

→ **5 openUrls(urls)**
only if files were named: EditorController::instance()-> openUrls(urls) opens them in tabs

→ **6 QGuiApplication::exec()**
the event loop: the app runs until you quit.
The window shows by itself: Main.qml says visible: true.

How the controllers depend on each other at start-up

An arrow means "its constructor needs". So EditorController always finishes first, whatever order QML picks.



EditorController's own constructor needs no other controller. It makes one empty tab, the input state, the row builder and the rows model, then calls rebuild(true). In the measured run, MinimapController started first and created it.

ShortcutsController needs nobody: it reads its keys from QSettings. EditorController only reaches it on the first key press.

Each controller makes the link itself, in its constructor, so main.cpp needs to know only EditorController.

Building, packaging, releasing

One platform in practice: Linux x86_64. Three ways from source to a running app, all starting from CMakeLists.txt.

CMakeLists.txt

textengine
static library: every .cpp of backend/textengine, no Qt

appChaoticNotepad
the program: main.cpp

QML module ChaoticNotepad
Main.qml, frontend/, wrapper/, the icon font

links
Qt6::Quick, Qt6::Network and textengine

C++17, Qt 6.10 or newer, version 2.3.2

install.sh
sudo: from source, into /usr/local

cmake Release, build in build/
cmake --install
refreshes desktop and icon caches

packaging/build-appimage.sh
on your computer, with a version

builds in build-release/, installs to AppDir/
linuxdeploy collects Qt, then the original
libraries are put back; writes dist/

.github/workflows/release.yml
on GitHub, for a pushed v* tag

Ubuntu 22.04, Qt 6.11.2
runs build-appimage.sh, smoke test
publishes the release

Making a release

ordinary pushes build nothing: only a v* tag starts this

git push origin v2.3.2
a tag, not a commit

→ **install packages**
and Qt 6.11.2

→ **work out the version**
v2.3.2 gives 2.3.2

→ **build-appimage.sh**
AppImage, tar.xz, checksums

→ **sign-update.sh**
no key: stops here

→ **smoke test**
run 8 s under xvfb

→ **upload run files**
kept with the run

→ **publish release**
only for a tag

The smoke test does not stop a release: it is marked continue-on-error, so after a crash at start-up the run goes on and publishing still happens. "Run workflow" on the Actions page builds and keeps the files, and publishes nothing.

What a release contains

ChaoticNotepad-2.3.2-x86_64.AppImage
one file with Qt inside: run it

ChaoticNotepad-2.3.2-linux-x86_64.tar.xz
the app and scripts: install-app.sh, no root, in your home

SHA256SUMS.txt, update-manifest.txt and its .sig
checksums, and the signed list the updater checks

How each install is removed

install.sh
sudo ./uninstall.sh removes it from /usr/local

install-app.sh
./uninstall-app.sh removes it from your home; documents and settings stay

AppImage
nothing is installed: delete the file

Platforms and tools

The release workflow and packaging scripts are Linux only, and there are no per-platform source files. CMakeLists.txt keeps Qt's macOS and Windows settings from the project template, and fileio.cpp has Unix-only code to flush a save to disk, but no Windows or macOS build is made or tested here.

Why the script puts libraries back: the AppImage tools rewrite the headers of what they bundle, which crashed the app at start-up on Fedora 44. AppRun then sets the library paths itself.

build-appimage.sh needs Qt 6 with qmake, cmake, curl and a C++17 compiler. It downloads linuxdeploy, its Qt plugin and appimagetool once, into tools/. Generated folders: build-release/, AppDir/, dist/.

Updating and closing

Nothing updates by itself: the app only tells you. Every step that can go wrong leaves the old version working.

1 Check

once a day, if the switch is on

GET releases/latest

newer: green dot in the toolbar

offline: says nothing

2 Verify

you click Update now

manifest and its signature first

Ed25519, public key built in

version must be newer

3 Download

next to the program

tarball or AppImage

SHA-256 must match the manifest

tarball unpacked with tar

4 Quit

the normal close questions

Cancel: nothing is installed

other windows open: refused

old app leaves old-exited

5 Swap

install.sh takes over

old copy becomes previous

new copy takes its place

starts it, reopens your files

6 Confirm

the new version answers

after 2 s, or a clean quit: started

script deletes the backup

download already gone

What stops a bad update

Wrong key, edited manifest
the signature check fails: nothing is downloaded

Changed file
its SHA-256 differs from the signed manifest: deleted

An old release passed off as new
the signed version must be newer than this one

A name pointing somewhere else
the file names must match the version exactly

A stalled server
requests give up after 30 seconds

When the new version fails

install.sh gives the new version one minute to write the started file, and it only writes that after running for 2 seconds. If it exits, crashes or hangs, the script stops its whole process group, moves previous back, starts the old version and leaves a .failed mark. The old version then shows a message box.

A second window started during this sees the installing mark and leaves the folder alone. A shared lock on instances.lock tells the app when other windows are open: while they are, no update starts and no leftovers are cleaned up.

Who can update

`Tarball`
in ~/.local/share/ChaoticNotepad: the folder is swapped

`AppImage`
APPIMAGE and APPDIR must match where this program runs, so another app's leftover variables can never make it overwrite that app

`sudo ./install.sh`
needs root: gets the dot and a link to the release page

Where the key is

The private key signs a release in the workflow, from the secrets UPDATE_SIGNING_KEY_BASE64 and UPDATE_SIGNING_KEY_PASSWORD. It is never in the repository. The public key is in updatecontroller.cpp and the README.

Closing the app

DialogController goes through the unsaved tabs one at a time: it shows the tab, then asks Don't save, Cancel or Save. Cancel, a failed save or a cancelled Save As stops quitting. The update's restart uses this same path.